Prabhavit Gupta

Software / Machine Learning Engineer: Python, Data Pipelines & Model Systems

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Summary

Software / Machine Learning Engineer with 3 years of commercial production experience architecting scalable Python backend systems, automated data extraction pipelines, and machine learning workflows. Combines rigorous software engineering practices—clean architecture, automated testing, and containerized CI/CD deployments—with practical experience integrating modern ML and generative models (Hugging Face, embeddings, OpenAI, Anthropic). Demonstrated capability building high-throughput relational data workflows in PostgreSQL, designing deterministic validation rules over unstructured documents, and maintaining observable cloud microservices on AWS and Linux.

Experience

FlowSpace.co • *Lead Developer & AI Systems Architect (Freelance)*
Mar 2024 - Aug 2026 • London, UK (remote)

- Engineered production Python backend services and data transformation pipelines processing unstructured multi-format records (invoices, forms, identity documents) for 4,500+ active enterprise users.
- Implemented structured schema validation layers and deterministic fallback guards on model inference outputs, preventing malformed or invalid records from entering production databases.
- Optimized PostgreSQL schemas, queries, and indexing strategies for high-volume tabular records and transactional metadata across distributed customer accounts.
- Built offline benchmark evaluation harnesses using golden ground-truth datasets to test model accuracy, feature drift, and regression risks prior to production deployment.
- Integrated asynchronous worker queues with exponential backoff retries and idempotency keys to ensure zero-loss processing during high-traffic load spikes.
- Automated continuous integration, static analysis, and unit test suites via GitHub Actions and Docker containerization.
- *Tools: Python, PostgreSQL, Hugging Face, REST APIs, Docker, AWS, Supabase, Linux*

Urlink • *Project Manager & Solutions Developer (Freelance)*
Apr 2025 - Jul 2025 • Barcelona, Spain (remote)

- Developed automated Python data ingestion scripts and entity transformation routines connecting external REST endpoints to centralized PostgreSQL tables.
- Designed automated data sanitization and schema consistency checks, maintaining high data integrity across multi-tenant enterprise accounts.
- Collaborated across product and engineering stakeholders to define interface contracts, technical requirements, and system monitoring alerts.
- *Tools: Python, REST APIs, PostgreSQL, Supabase, TypeScript*

Independent Technical Consultant • *AI Automation & Integration Developer*
Sep 2023 - Mar 2024 • Venice, Italy

- Developed Python automation workers and webhook handlers for automated document parsing, data enrichment, and database ingestion.
- Built automated regression test suites and end-to-end integration monitors to catch breaking API and schema changes early.
- Documented data schemas, operational runbooks, and API contracts for cross-functional client teams.
- *Tools: Python, Webhooks, REST APIs, Supabase, Linux*

Tractionboard.co • *Low-Code Developer (Internship)*
Aug 2023 - Oct 2023 • Utrecht, Netherlands (remote)

- Integrated machine learning APIs and email notification services, building automated validation pipelines for core user workflows.
- *Tools: Python, OpenAI API, Sendgrid, REST APIs*

Education

Ca' Foscari University of Venice • *Bachelor's in Philosophy, International Studies & Economics, Minor in Entrepreneurship*
Oct 2020 - Jun 2024 • Venice, Italy

Certifications

Microsoft: Introduction to building AI agents • Microsoft Copilot Studio Agent Academy • Microsoft Power Automate • Tray.AI Build Practitioner (Sep 2024) • Tray.AI Build Foundations

Additional Information

- **Work authorization:** European Long Term Residence Permit (Italy), unrestricted EU work authorization (Partita IVA / Italian employment ready). No sponsorship required.
- **Languages:** English (fluent / C1 professional), Hindi (native), Italian (beginner).
- **Availability:** Available immediately for full-time remote roles in Italy.

Core Skills

- **Software Engineering & Python:** Python 3, clean code principles, modular software architecture, object-oriented design, error handling, unit/integration testing (pytest).
- **Machine Learning & Generative AI:** Hugging Face, PyTorch, embeddings, vector similarity search, LLMs (OpenAI, Anthropic Claude), model evaluation harnesses, prompt optimization.
- **Data Systems & Databases:** PostgreSQL, relational schema design, SQL query optimization, data cleansing and normalization, Supabase, Redis caching.
- **Cloud & DevOps Practices:** Docker containerization, AWS cloud environments, Linux system administration, Git, GitHub Actions CI/CD, asynchronous queues.
- **Quality & Reliability:** End-to-end regression testing, golden dataset benchmarking, schema validation, telemetry and performance monitoring.